

Ekjot Brar

780-904-0343 | ekjotbrar110@gmail.com | linkedin.com/in/ekjot-brar/ | github.com/EkjotBrar | ekjotbrar.com

EDUCATION

University of Alberta

Bachelor of Applied Science, Minor in Applied Mathematics

Edmonton, AB

Sep. 2023 – May 2028

EXPERIENCE

Artificial Intelligence Research Assistant

May 2026 – August 2026

University of Alberta

Edmonton, AB

- Researched neuro-fuzzy and machine learning techniques for intelligent decision-making systems, performing model development, testing, and performance analysis.
- Developed Python-based data processing and evaluation pipelines, supporting reproducible experimentation and quantitative assessment of AI algorithms.

Service Router Platform Test Dev

Jan. 2026 – May 2026

Nokia

Ottawa, ON

- Engineered and scaled modular SR/SR-L router topologies, improving re-usability and efficiency of network validation frameworks.
- Developed Python-based automation pipelines in a Linux (Ubuntu) regression environment for carrier-grade router hardware/software validation.
- Built dynamic Precision Time Protocol (PTP) test environments with parameterized configurations (IP, VLAN, interface speeds) to accelerate automated test setup.
- Conducted system-level validation of synchronization technologies (PTP, GNSS, SyncE), ensuring accurate timing and clock alignment across network devices.
- Designed object-oriented Python abstractions mapped to YANG data models for real-time configuration and monitoring of central frequency clock systems.

CAN. Technologist

May 2025 – August 2025

Tetra Tech

Edmonton, AB

- Supported field deployment and troubleshooting of advanced pavement inspection systems (LiDAR, LCMS, laser sensors) ensuring reliable high-resolution data collection.
- Developed a QR-based tracking system for hard drives using structured database workflows to improve data traceability and inventory management.
- Processed and validated pavement datasets using MATLAB, identifying anomalies (bridges, bumps) and improving data quality assurance workflows.
- Developed C/C++-based tools for configuration and data handling of imaging systems (Ladybug camera), supporting integration into analysis pipelines.

PROJECTS

ECO Car | Python, C/C++, KiCAD, STM32IDE

May 2025 – Present

- Contributed to Shell Eco-Marathon hydrogen vehicle development, gaining hands-on experience with embedded systems and low-power electronics relevant to semiconductor applications.
- Designed and routed a 2-layer PCB in KiCad, including schematic capture, component selection, and layout optimization for efficient electronic system integration.
- Programmed an STM32 microcontroller for real-time control and PWM-based LED operation, building embedded C and digital control experience.
- Assembled and soldered SMD/through-hole components, supporting hardware prototyping and circuit validation.
- Implemented FPGA digital logic using VHDL, including finite state machines and combinational/sequential circuit design.

AlbertaSAT | ANSYS Mechanical, SolidWorks, AutoCAD, Git, MATLAB

May 2018 – May 2020

- Contributed to the development and microgravity testing of a gecko-adhesive debris capture system (CAN-RGX), focusing on system-level performance validation under dynamic and structural constraints
- Performed vibrational and structural analysis of CubeSat subsystems, evaluating modal response, stress distribution, and mechanical stability of critical components
- Supported Ex-Alta 3 CubeSat hardware development, including hinge mechanism design, CAD modeling, and investigation of UHF integration with emphasis on mechanical-electrical system reliability

TECHNICAL SKILLS

Languages: C/C++, Python, MATLAB, Java, SQL, HTML/CSS, R

Embedded & Hardware: STM32, FPGA, VHDL, KiCAD, Altium, Oscilloscope, Logic Analyzer

Developer Tools: Linux (Ubuntu), Git, VS Code, Visual Studio, MATLAB

Design & Simulation: KiCad, SolidWorks, ANSYS Mechanical, AutoCAD